PROJECT BRIEF

Team Name: JARDS, PROJECT TITLE: NEST HELPER

1. Problem Statement: New parents suffer from tiredness and a heavy mental load. Replenishing consumables (diapers, wipes) and updating baby clothes is reactive, stressful, and chaotic, leading to midnight emergencies and lapsed customers. Existing retailers rely on generic, spammy discount email blasts rather than using the rich transactional and age-band data they already hold to serve parents proactively.

2.**Target Users:** Our primary users are time-poor parents and caregivers of babies up to 3 years who face constant mental load from work and raising a baby and require frictionless, spam-free restocking of daily essentials alongside timely sizing and milestone guidance as their baby rapidly grows.

3**.Proposed Solution: Nest Helper**

Nest Helper is a conversational commerce assistant which models the two biological schedules that govern infant care:
The Depletion Schedule: It uses each family's own reorder gap to predict when a pack runs out and offers a reorder around then
The Growth Schedule: It maps developmental age bands to anticipate physical growth transitions—proactively recommending clothing size-ups, diaper stage upgrades, and Stage Bundles.

4. Core Workflow & Priority Engine

Phase / Priority	Core Function	Action & Guardrails
Ingestion & Profiling	Data Analysis	Reads order history, baby birth month and consent to estimate sizing needs and when products run out.
Priority 1: Refilling	Consumables Auto-Prompt	Picks a refill reminder when consumables (diapers, wipes, Stage 3 growing-up milk) are predicted to run low.
Priority 2: Size Up	Growth Transition	Recommends size transitions for clothing and diapers based on infant growth timelines.
Priority 3: Milestones	Milestone Offer	At 6 months and the first birthday: free delivery on the next order within 7 days.
Priority 4: Engagement	Re-activation	Re-engages inactive customers with free delivery within 14 days for families with 2+ orders and 60+ days quiet.
Responsible AI Filter	Compliance Guardrails	Enforces strict opt-ins. Strictly prohibits bot-initiated prompts, discounts, or bundles for Stage 1 & 2 infant formula (parents must initiate formula reorders independently).
Escalation	Switch to live chat	For medical questions the AI gives no advice and tells the parent to type HUMAN, then tap "Connect me now" to reach a team member (Mon–Fri, 9am–6pm).

5. Tools & Technology Used:

- Conversational Interface: MantaGO Platform
- Cloud & DevOps: Render.com (API Hosting & SSL), GitHub & GitHub Pages (E-Commerce Storefront).
- Engine & AI: Python 3 (decision engine, message checker); Google Gemini 3.1 Flash-Lite inside MantaGO (AI answers).

6. Expected Impact

- Higher Retention and Repeat Purchases: Timely replenishment and milestone prompts should raise repeat purchases and LTV.
- Frictionless Parent Experience: Reorder buttons that open the right product, free size exchange offers, and relevant bundle suggestions save parents time and reduce decision fatigue.
- Ethical & Trustworthy Brand Image: Consent, a 7-day message limit, STOP and strict compliance with marketing restrictions build long-term parental trust.

7. Current Limitations & Next Steps

- Limitations: Age and depletion estimates rely on historical order heuristics (skewed by gift purchases or multi-child households); shop, customers and results are simulated; our MantaGO plan cannot send messages on a schedule, so reminders are shown through the DEMO flow.
- Next Steps: Daily scheduled sending; a real "Remind me in 3 days" snooze; Integrate live payment/checkout APIs for true one-tap purchasing; implement machine learning to dynamically refine consumption rate predictions; expand multi-child profile support; and run A/B testing on message packs to optimize conversion rates.